

Part A Chapter 7

Emotion Contagion¹

1 Introduction

Within psychology, emotion is often defined as a state or process that plays a role in various cognitive processes, among which decision making and action preparation. Emotions are elicited by a particular stimulus, often include physiological reactions, and are relatively intense and short-lived. Emotions also have a social function: humans may experience



situations where expressions of emotion by one individual shape the emotions, thoughts and behaviour of others; others' reactions can then influence their future interactions with the individual expressing the original emotion, as well as the person's future emotions and behaviours. This every day, continuous, automatic process of emotion contagion has been described as a tendency to mimic the nonverbal behaviour of others, to synchronize facial expressions, vocalizations, postures, and movements with others, in order to converge emotionally..

2 Analysis of the Main Aspects and Their Relations

2.1 Identification of relevant concepts

Modelling group emotion can be done at the level of the group or at the level of the individuals, which can be named respectively the top-down and bottom-up approach. The bottom-up perspective sees group emotion as the sum of its parts, affected by the homogeneity or heterogeneity of the group and the mean emotions of the group members. Individual differences play an important role, such as specific personality traits and the

¹ Based on:

Bosse, T., Duell, R., Memon, Z.A., Treur, J., and Wal, C.N. van der (2009), A Multi-Agent Model for Mutual Absorption of Emotions. In: Otamendi, J., Bargiela, A., Montes, J.L., Pedrera, L.M.D. (eds.), Proceedings of the 23th European Conference on Modelling and Simulation, ECMS'09. European Council on Modeling and Simulation, 2009, pp. 212-218.

Duell, R., Memon, Z.A., Treur, J., and Wal, C.N. van der (2009), An Ambient Agent Model for Group Emotion Support. In: Cohn, J., Nijholt, A., Pantic, M.(eds.), Proc. of the Third Int. Conference on Affective Computing and Intelligent Interaction, ACII'09. IEEE Computer Society Press, 2009, pp. 550-557..

underlying brain mechanisms. There is consensus that the basic structure of personality incorporates five super ordinate factors: extraversion, agreeableness, conscientiousness, emotionality and intellect. Extraversion is characterized as a tendency towards sociability, activeness, dominance and craving excitement. Agreeableness is thought of as being concerned with the maintaining of relationships. Conscientiousness reflects qualities of planning, persistence, and purposeful striving towards goals. Emotionality (or neuroticism) concerns the ease and frequency with which a person becomes upset and distressed. Intellect (or openness) reflects openness to experience and intelligence-related sub-traits such as intellect, creativity and curiosity. Personality dimensions have been linked to specific aspects of brain functioning. The functional approach considers two sets of structures in the brain, the behavioural approach system (BAS), and behavioural inhibition system (BIS). BAS causes animals and humans to move towards incentives: things they desire. Besides managing the approach, BAS also creates excitement and positive feelings. The neurotransmitter linked to BAS is dopamine. The system BIS can be thought of as a stopping system, a system responsive to threat. BIS is responsive to cues of punishment and danger, not incentives. The neurotransmitter linked to BIS is serotonin. The feeling BIS creates is anxiety. The two systems have been linked with personality traits: BIS is critical to neuroticism and BAS to extraversion (seeking social incentives).

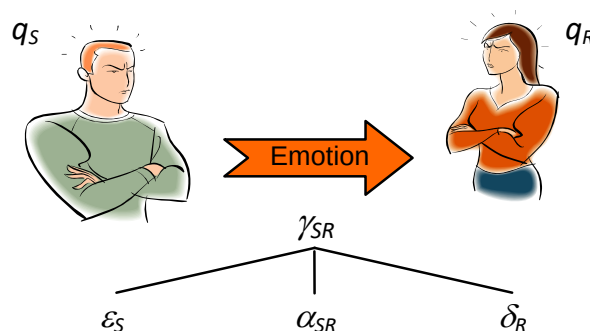


Figure 1 Aspects of emotion contagion

Personality also has been related to the functioning of the nervous system in a variable called sensation seeking. Persons who score high on sensation seeking are in search for new and exciting experiences. They are more likely to do high-risk sports, such as skydiving, are more sexually experienced and sexually responsive and compared to people low on this trait, are faster drivers. It has been suggested that there is a particular brain chemical for sensation seeking (an enzyme called monoamine oxidase, MAO), which regulates several neurotransmitters, including serotonin and dopamine. Sensation seeking has also been related to the previously mentioned BIS and BAS systems. Persons high on sensation seeking correspond to a high BAS and low (sensitive) BIS.

The model introduced in this chapter has been designed as an interpretation of the bottom-up approach where group emotion can be seen as the sum of its parts. Here the parts are the emotion levels q_A of individual group members A . The model distinguishes multiple factors

that influence emotion contagion processes. The model incorporates individual differences in personality traits: neuroticism and extraversion (BIS and BAS). A number of aspects of the proposed computational model are distinguished that play a role in the contagion, varying from characteristics related to an individual S sending the emotion (expressiveness ε_S), an individual R receiving the emotion (openness δ_R), and the channel between sender S and receiver R of the emotion (channel strength α_{SR}); see Figure 1.

The aspect ε_S describes how introvert or extravert, expressive, and/or active or energetic the person is. This aspect relates to the personality trait extraversion. It represents in how far a person transforms internal emotion into external expression. For example, an introvert person will induce a weaker contagion of an emotion than an extravert person. Notice that the level of emotion and the strength by which it is expressed and contaged are independent concepts. For example, it is possible to only weakly express and contage a strong feeling of sadness, or, the other way around, a slight feeling of sadness can be strongly expressed and contaged.

The aspect α_{SR} depends on the type and intensity of the contact between the two persons (e.g., distance vs. attachment). The aspect δ_R indicates the degree of susceptibility or openness of the receiver. This represents in how far the receiver allows the emotions received from others to affect the own emotion, and how flexible/persistent the person is emotionally. The parameter α_{SR} may be related to a combination of more specific aspects such as the directness of the emotion contagion, and the relations between sender and receiver. Emotion contagion is direct if the persons infecting each other with emotions are together in the same room and pay attention to each other. Indirect contagion can happen when for instance the contagion between others is observed. Direct contagion is propagated stronger than indirect emotion contagion. The aspect γ_{SR} is used to represent the *contagion strength from S to R* by which an emotion is received by R from sender S , considered as the combination of the different steps based on the aspects ε_S , α_{SR} , δ_R . In a way γ_{SR} expresses the energy level with which an emotion is being expressed, transferred and received.

Based on the emotion levels of the individual group members a level of the *group's emotion state* can be determined according to a certain measure. Within this measure the emotion levels of specific group members will contribute, but maybe not for each member in a comparable manner. Therefore for each group member A a (relative) *relevance factor* ρ_A is considered indicating to which extent this member is relevant for the measure of the group's emotion state.

In summary, the following concepts play a role in emotion contagion:

- a person's emotion expressiveness (ε_A)
- a person's openness for received emotion (δ_A)
- the strength of the channel from sender to receiver (α_{BA})
- the contagion strength from a sender to a receiver (γ_{BA})
- the level of a person's emotion (q_A)

- a person's relevance for the level of the group's emotion state (ρ_A)
- the level of the group's emotion state (q)

The global dynamic relationships are as follows:

Affecting emotion level (Figure 2 and Figure 3)

- the level of a person's emotion affects the level of another person's emotion
- the contagion strength from a sender to a receiver affects the level of the receiver's emotion

Affecting contagion strength (Figure 2 and Figure 3)

- a sender's emotion expressiveness affects the contagion strength from this sender to any receiver
- a receiver's openness for received emotion affects the contagion strength from any sender to this receiver
- the strength of the channel from a sender to a receiver affects the contagion strength from this sender to this receiver

Affecting group emotion state (Figure 4)

- the level of a person's emotion affects the level of the group's emotion state
- a person's relevance factor affects the level of the group's emotion state

In Figure 2 the dynamic relationships for the emotion transferred to person $a3$ in a group of three persons $\{a1, a2, a3\}$ are depicted.

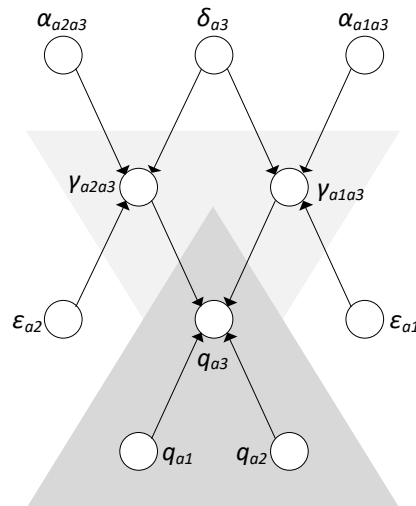


Figure 2 Emotion contagion received by person $a3$ within a group of three persons

A similar picture can be made for each of the other two persons in the group: $a1$ and $a2$. A picture of the dynamic relationships for emotion contagion in the whole group can be obtained by combining the three pictures for each of the three persons, as shown in Figure 3.

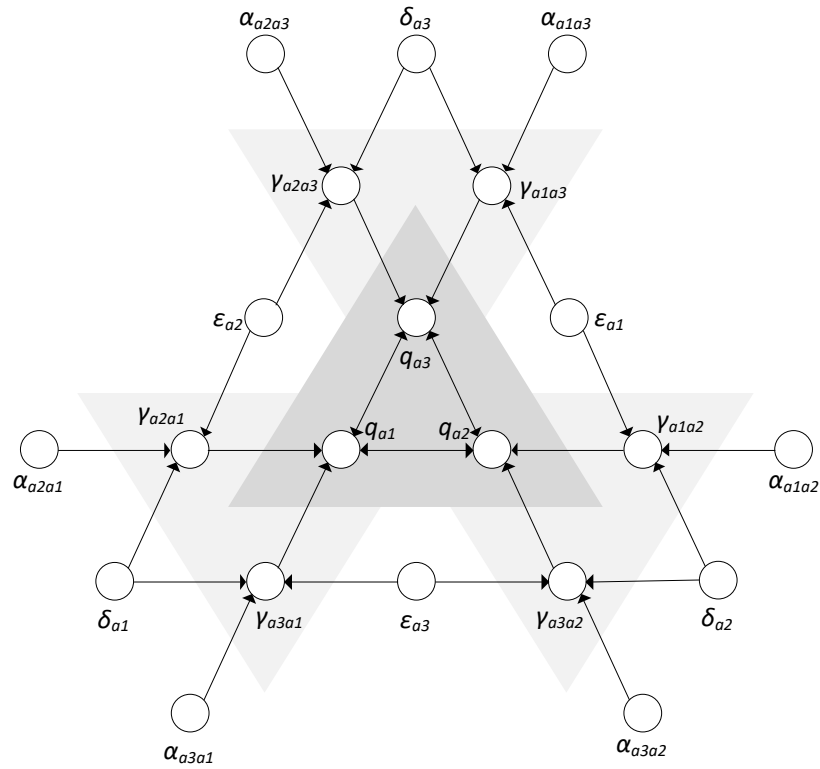


Figure 3 Dynamics relationships for emotion contagion between members within a group of three persons

In addition to the process shown in Figure 3 to determine the group member's emotion levels, Figure 4 depicts the dynamic relationships to determine the group's emotion level from the individual emotion levels of the group members.

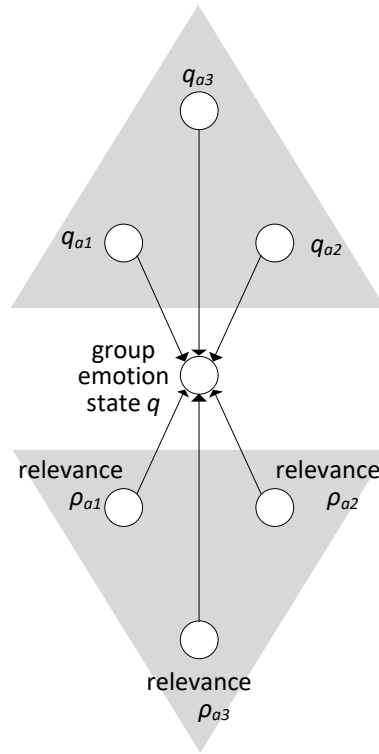


Figure 4 Relationships for the group's emotion level in a group of three persons

The Detailed Domain Model

The model has been formalised in hybrid manner, by embedding quantitative, numerical representations within qualitative, logical representations. The numerical formalisation represents the different aspects by numbers in the interval $[0, 1]$; see Table 1.

emotion expressiveness for sent emotion	ε_S
openness for received emotion	δ_R
the strength of the channel from sender to receiver	α_{SR}
contagion strength from sender to receiver	γ_{SR}
overall contagion strength for receiver	γ_R
relevance factor for group member A	ρ_A
emotion level of group member A	q_A
level of the group's emotion state	q

Table 1 Numerical formalisation of concepts related to a sender S, receiver R, or both

The logical formalisation incorporating the numerical representations, is shown in Table 2; Table 3 shows the sorts used.

Concept	Formalisation
person B has emotion expressiveness E	<code>has_expressiveness(B:AGENT, E:REAL)</code>
person A has openness for received emotion D	<code>has_openness(A:AGENT, D:REAL)</code>
channel from sender B to receiver A has strength C	<code>has_channel_strength(B:AGENT, A:AGENT, C:REAL)</code>
contagion strength from B to A is CS	<code>has_contagion_strength(B:AGENT, A:AGENT, CS:REAL)</code>
overall contagion strength to receiver A is CS	<code>has_overall_contagion_strength(A:AGENT, CS:REAL)</code>
person A has relevance factor R	<code>has_relevance(A:AGENT, R:REAL)</code>
person A has emotion level V	<code>has_emotion_level(A:AGENT, V:REAL)</code>
the group has emotion level V	<code>group_emotion_level(V:REAL)</code>

Table 2 Logical formalisation of the different concepts

Sort	Description of use	Elements
AGENT	group members	<code>a1, a2, a3, a, b, c, arnie, bernie, charlie</code>
REAL	group member emotion level, group emotion level, expressiveness factor, channel strength, openness, contagion strength, relevance factor	<code>real numbers</code>

Table 3 Sorts used

The stronger the expressiveness, the channel, and the openness, the higher ε_S , α_{SR} , and δ_R and the more contagion will take place. The parameter γ_{SR} used to represent the contagion strength from S to R is modelled as the combination of the different steps by multiplying them:

$$\gamma_{SR} = \varepsilon_S \alpha_{SR} \delta_R$$

In a way γ_{SR} expresses the energy level with which an emotion is being expressed, transferred and received. The idea is that each of these numbers indicates the fraction of strength remaining after, respectively, expression, transfer and receipt. For example, when $\varepsilon_S = 0.9$, $\alpha_{SR} = 1$, and $\delta_R = 0.9$, then after expression the emotion strength is reduced to 0.9 , and after transfer and receipt to 0.81 . If γ_{SR} is 0 , there will be no contagion from S to R , if it is 1 , there will be a maximum strength of contagion from S to R . If γ_{SR} is not 0 , there will be contagion and the higher the value, the more contagion will take place. The *overall contagion strength* by which emotions from all the other group members are received by A , indicated by γ_A , is defined as the sum of $\gamma_{BA} = \varepsilon_B \alpha_{BA} \delta_A$ for all other persons, i.e.,

$$\gamma_A = \sum_{B \neq A} \gamma_{BA}$$

Note that, as all terms γ_{BA} are between 0 and 1 , also depending on the size n of the group, γ_A can be higher than 1 , up to $n-1$.

The general idea behind the detailed model is as follows. At each time point, for each group member A :

- Determine the average emotion impact q_A^* of the other group members, taking into account that not all members have an equal impact on A .
- Determine the difference $q_A^* - q_A$ of this average q_A^* with the current emotion level q_A of person A .
- Adjust the emotion level q_A by a fraction γ of this difference, so that the new emotion level becomes $q_A + \gamma (q_A^* - q_A)$; another way of expressing this is that the new emotion level of A becomes a weighted average of the old emotion level q_A and the average value q_A^* of the other group members: $(1-\gamma) q_A + \gamma q_A^*$.

To apply this idea, the main questions to be answered are: (1) how is the average emotion impact q_A^* of the emotion levels of the other persons determined, and (2) which fraction γ is used to adjust the own emotion level. An answer for (1) is obtained by taking a weighted sum of the emotion levels q_B for $B \neq A$ with weights w_{BA} :

$$q_A^* = \sum_{B \neq A} w_{BA} q_B$$

Here the weights w_{BA} are taken proportional to the impact of B on A as described by the contagion strength γ_{BA} . By a normalisation to achieve that their sum is 1, the weights are defined by

$$w_{BA} = \gamma_{BA} / \gamma_A$$

so

$$q_A^* = \sum_{B \neq A} \gamma_{BA} q_B / \gamma_A$$

An answer to (2) can be obtained by using the product of overall contagion strength and step size $\gamma_A \Delta t$ as the fraction γ for A . As a result, the model for emotion contagion is expressed in numerical representation by:

$$q_A(t+\Delta t) = q_A(t) + \gamma_A (q_A^* - q_A) \Delta t$$

for all group members A . Here for direct simulation the step size Δt should be taken sufficiently small: at most $1/(n-1)$ when n is the group size (otherwise the factor $\gamma_A \Delta t$ may be higher than 1 and the emotion levels may run outside the interval $[0, 1]$). Note that alternatively the numerical model equations can be rewritten as:

$$\begin{aligned} q_A(t+\Delta t) &= q_A(t) + (\gamma_A q_A^* - \gamma_A q_A) \Delta t \\ &= q_A(t) + (\sum_{B \neq A} \gamma_{BA} q_B - \sum_{B \neq A} \gamma_{BA} q_A) \Delta t \\ &= q_A(t) + \sum_{B \neq A} \gamma_{BA} (q_B - q_A) \Delta t \end{aligned}$$

From this (equivalent) numerical representation it can be seen how all of the other persons B have their effect $\gamma_{BA}(q_B - q_A) \Delta t$ on the adjustment of q_A . When q_B is higher than q_A , this implies an increase of q_A , and when q_B is lower than q_A this implies a decrease of q_A . For example, when $\gamma_{BA} = 1$, $q_B = 0.9$, $q_A = 0.3$ and $\Delta t = 0.1$, then $\gamma_{BA}(q_B - q_A) \Delta t = 0.06$. Therefore in principle (when no other persons have effects on A) the emotion level of A will

become closer to that of B . Conversely, A has as effect $\gamma_{AB}(q_A - q_B)$ on the adjustment of B 's emotion level. This implies that in principle B 's emotion level will become closer to that of A . Note that in the symmetric special case that $\gamma_{AB} = \gamma_{BA}$ these effects have the same amount, but in opposite direction.

For a logical representation in semiformal and formal textual format the dynamic relationships for individual emotion levels were expressed as follows (for a group of three persons):

DDR1 Determining contagion strengths

If B has expressiveness E
 and the channel from B to A has strength C
 and A has openness D
 then the contagion strength from B to A will be $E \cdot C \cdot D$

```
has_expressiveness(B, E) &
has_channel_strength(B, A, C) &
has_openness(A, D)
→ has_contagion_strength(B, A, E * C * D)
```

DDR2 Updating emotion levels

If $A \neq B$ and $B \neq C$ and $C \neq A$
 and A has emotion level $V1$
 and B has emotion level $V2$
 and C has emotion level $V3$
 and the contagion strength from B to A is $CS2$
 and the contagion strength from C to A is $CS3$
 and the step size is DT
 then the emotion level of A will be $V1 + CS2 \cdot (V2 - V1) \cdot DT + CS3 \cdot (V3 - V1) \cdot DT$

```
A ≠ B & B ≠ C & C ≠ A &
has_emotion_level(A, V1) &
has_emotion_level(B, V2) &
has_emotion_level(C, V3) &
has_contagion_strength(B, A, CS2) &
has_contagion_strength(C, A, CS3) &
step_size(DT)
→ has_emotion_level(A, V1 + CS2 * (V2 - V1) * DT + CS3 * (V3 - V1) * DT)
```

How from the individual emotion levels, the level of the group's emotion state is determined, was formalised as follows:

DDR3 Determining the group's emotion level

If the group members have emotion levels $V1, V2, V3$
 and relevance factors $R1, R2, R3$ respectively
 then the group's emotion level is $R1 \cdot V1 + R2 \cdot V2 + R3 \cdot V3$

```
has_emotion_level(a1, V1) &
has_emotion_level(a2, V2) &
has_emotion_level(a3, V3) &
has_relevance(a1, R1) &
has_relevance(a2, R2) &
has_relevance(a3, R3)
→ group_emotion_level(R1 * V1 + R2 * V2 + R3 * V3)
```

Example Simulation Traces for the Domain Model

A large number of simulations have been performed, resulting in a variety of interesting patterns. In this section some of the simulation results are discussed. Simulations shown here are for a group of 3 agents: a, b, and c.

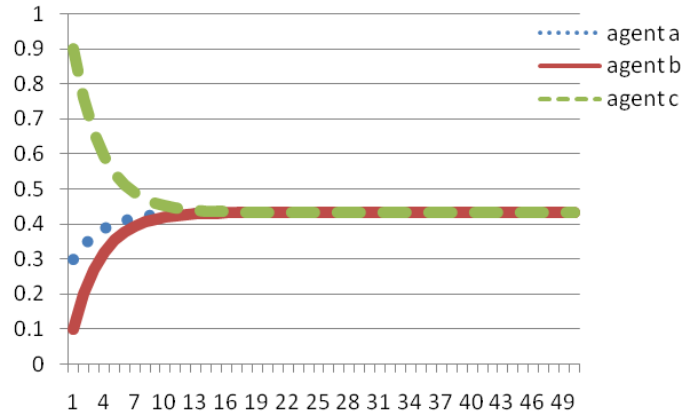


Figure 5 Simulation trace 1 (all $\gamma_A = 1$)

A first pattern found is that when the γ_A for all group members are not 0, in this case they are all 1, the emotion levels of all of them will approximate their average initial emotion level, with speed depending on the δ_A (susceptibility) and $\varepsilon_B \alpha_{BA}$; see Figure 5. The opposite of this pattern happens when all γ_A are 0, then all group members will have equilibria that are equal to their initial emotional levels. In other words: the emotional levels of all group members will not change at all. Another situation (see Figure 6) occurs when a has δ_a set to 0 and the other group members have this parameter $\neq 0$. This situation represents that a is not open to receive emotions, but can send emotions. As a result a 's initial emotion level will remain the same. Furthermore, b and c will eventually reach the equilibrium of a , which is equal to his initial emotion level 0.3.

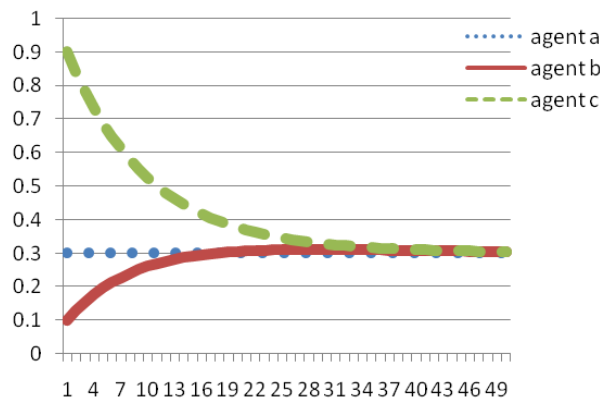


Figure 6 Simulation trace 2 ($\delta_a = 0$)

In Figure 7 it is shown that when both a and b have δ_A set to 0, person c will reach a value in between a and b 's initial emotion values. The actual value c reaches depends on the parameter settings for all group members. This situation represents a case where two group members do

not change their emotional level because they are only open to sending emotions, but not receiving emotions.

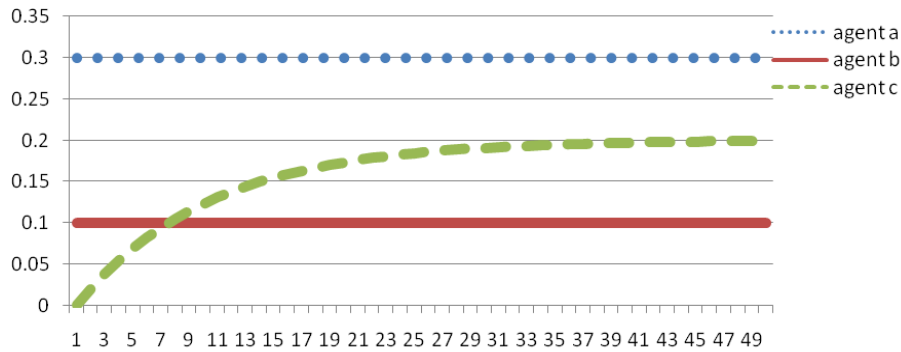


Figure 0 Simulation trace 3 ($\delta_r(a, b, c) = (0, 0, 0.5)$)

As a result the third person is forced to reach a value in between the emotional levels of the others. A next situation (see Figure 7) is one where δ_a and ε_a are set to 0. This represents person *a* being bidirectionally excluded from emotion contagion: (s)he cannot receive or send emotions. The persons *b* and *c* are forced to go to a certain average in between their initial emotion values. The exact value they reach depends on the settings of their δ_A (susceptibility) and $\varepsilon_B \alpha_{BA}$.

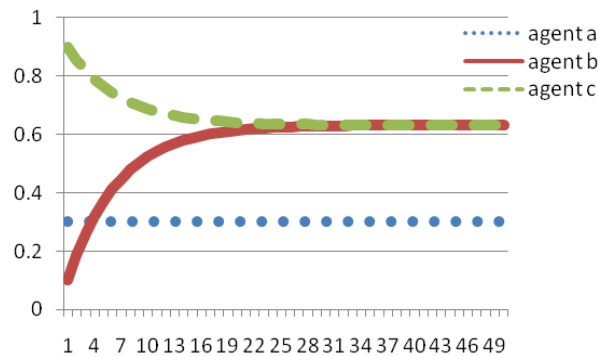


Figure 0 Simulation trace 4 ($\delta_a = \varepsilon_a = 0$)

References

- Ashby, W.R. (1952). *Design for a Brain*. Wiley and Sons, New York.
- Jonker, C.M., Lam, R.A., and Treur, J., (2001). A Reusable Multi-Agent Architecture for Active Intelligent Websites. *Journal of Applied Intelligence*, vol. 15, 2001, pp. 7-24.
- Kim, J. (1996). *Philosophy of Mind*. Westview Press.
- Maslin, K.T. (2001). *An Introduction to the Philosophy of Mind*. Polity Press/Blackwell Publishers, Cambridge, UK.
- Morgan, C.L. (1894). *An introduction to comparative psychology*. London: Scott, 1894.